

Summary Report - AEGIS_Cooling_Tower_Inspection_Test_Document.docx

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EXECUTIVE SUMMARY

A routine preventive maintenance inspection of Cooling Tower CT-04 on August 18, 2026, revealed several operational deviations exceeding internal action levels. These include elevated fan vibration, bearing temperature, and motor current, as well as a basin TDS exceeding the internal limit. Three damaged drift eliminator panels were identified and require replacement. Based on these recorded findings, immediate action is warranted.

KEY FINDINGS

The following deviations were recorded during the inspection of Cooling Tower CT-04:

- Fan assembly vibration: 7.2 mm/s RMS (above the 6.0 mm/s RMS internal action level).
- Fan bearing temperature: 86 °C (above the 80 °C internal action level).
- Motor current: 42 A (above the 38 A rated current).
- Basin TDS: 1850 ppm (above the 1500 ppm internal limit).
- Three drift eliminator panels were visibly damaged and require replacement.

DETAILED ANALYSIS

Cooling tower range was calculated as 36.8 °C - 31.5 °C. Cooling tower approach was calculated as 31.5 °C - 27.0 °C.

Motor current exceeded the rated current by $(42\text{ A} - 38\text{ A}) / 38\text{ A} * 100 = 10.53\%$. Fan vibration exceeded the action level by $7.2\text{ mm/s} - 6.0\text{ mm/s} = 1.2\text{ mm/s}$.

Basin TDS exceeded the internal limit by $1850\text{ ppm} - 1500\text{ ppm} = 350\text{ ppm}$. No visible structural cracking was observed.

RISKS AND OPERATIONAL ISSUES

The observed exceedances of action levels pose operational risks. Elevated fan bearing temperature could indicate premature bearing failure, leading to increased maintenance costs and potential downtime.

Above-rated motor current could result in motor overheating and potential damage. The elevated basin TDS suggests a problem with the water treatment system, potentially impacting cooling efficiency and the integrity of equipment due to scaling or corrosion. The damaged drift eliminator panels will reduce cooling tower efficiency.

RECOMMENDATIONS

Based on the recorded findings, the following actions are recommended:

1. **Priority 1:** Inspect fan alignment, balance, mounting, and bearing condition; record follow-up vibration and temperature readings.
2. **Priority 2:** Investigate the cause of motor current above the rated value and verify motor/load condition.
3. **Priority 3:** Review basin water treatment/blowdown condition because TDS exceeds the internal limit.
4. **Priority 4:** Replace the three damaged drift eliminator panels.

Follow-up: Repeat relevant measurements after maintenance and record the new values.

AUTHORITATIVE SOURCES & CITATIONS

Document Name	Referenced Pages	Relevance Status
AEGIS_Cooling_Tower_Inspection_Test_Document.docx	Page(s) 1, 2	High

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